

Cecilia Regueira Moreira

Washington DC, USA

ceciliaregueira.m@gmail.com | 515-441-6013 | [LinkedIn](#) | [Web page](#)

SUMMARY

Senior Data Scientist/ Analyst with 10+ years of experience delivering predictive analytics, machine learning models, and institution-wide forecasting for higher education leaders. Proven track record designing enrollment and financial models used by Provosts and executive teams to guide recruitment strategy, financial aid optimization, and long-term planning. Expert in time series forecasting, classification models, geographic analysis, and data warehouse development, with deep experience translating complex models into actionable decisions for non-technical stakeholders.

- Policy and causal analysis
- Statistical modeling
- Data management and integration
- Data visualization and storytelling
- Machine learning
- Project leadership

PROFESIONAL EXPERIENCE

Senior Data Analyst (Analyst II), Iowa State University– Ames, Iowa
August 2022 – July 2026

- Lead data scientist for institution-wide enrollment forecasting, developing and maintaining time-series models used by the Provost's office to inform budgeting, hiring, and strategic planning.
- Designed and deployed machine learning models to predict enrollment, student yield, retention, and performance, supporting decisions across admissions, recruitment, and student success.
- Built financial aid elasticity models to optimize net tuition revenue and evaluate trade-offs between aid allocation and enrollment outcomes.
- Developed geographic and market-level models to identify high-value recruitment regions and target prospective students more effectively.
- Partnered directly with enrollment leadership and recruitment teams to translate business questions into analytical frameworks and operational dashboards.
- Played a key role in the design and development of a new institutional data warehouse following the university's transition to Workday, including data modeling, validation, and integration.
- Developed fuzzy-matching and entity-resolution models to identify and resolve duplicate student records across systems.
- Built reproducible datasets, feature-engineering pipelines, and documented workflows using Python, SQL, and version-controlled environments.
- Presented model outputs and forecasts to senior leadership, translating uncertainty, assumptions, and trade-offs into decision-ready insights.
- Selected Modeling & Analytics Projects
 - Time-series enrollment forecasting (institution-wide)
 - Student yield & retention classification models
 - Financial aid elasticity and net tuition optimization
 - Geographic market segmentation for recruitment
 - Student record entity resolution using fuzzy matching
 - Data warehouse modeling during ERP transition (Workday)

Data Analyst, Corteva– Jhonston, Iowa
May 2021 to July 2022

- Supported business operations across Seed Production, Supply Chain, Quality Supply, Sales, and Information Technology divisions in North America, Latin America, and Europe.
- Served as lead coordinator for Spanish and Portuguese-speaking partners, aligning data requirements across regional stakeholders and facilitating data integration across markets.
- Led cross-functional efforts to improve master data quality, developing governance protocols, QA procedures, and automated pipelines to ensure reliable reporting.
- Partnered with marketing, engineering, product development, data security, and compliance teams to implement scalable data solutions and insights to increase operational efficiency.
- Delivered periodic reporting to senior leadership, identifying automation opportunities to increase report frequency, reduce manual processing, and improve data quality.

External consultant, Inter-American Development Bank
Dec 2020 to May 2021

- Conducted labor market impact analysis using historical administrative data from the Uruguayan Social Security Agency to assess job creation, layoffs, and employment dynamics across economic sectors during the COVID-19 pandemic.
- Performed extensive data wrangling using Python and SQL to clean, normalize, and structure multi-year employment records, enabling accurate cross-sector comparisons and trend analysis.
- Authored technical reports and developed key performance indicators to guide future monitoring and policy evaluation efforts.

Analyst & Data Scientist, Aurea Consulting – Montevideo, Uruguay
Sep 2020 to Feb 2021

- Lead research analyst on a project evaluating the impact of financial market development on productivity growth in local industries.
- Constructed a panel dataset of industries from 63 countries (1988–2015) using SQL and Python.
- Applied advanced statistical modeling techniques to analyze total factor productivity (TFP) growth.
- Delivered insights to national policymakers and international agencies to inform policy design.

Retail pricing behavior research (Dissertation project), Universidad de la República
Aug 2019 to Aug 2020

- Conducted economic research analyzing price-setting strategies among Uruguayan retail chains, focusing on market structure and coordination behavior.
- Led extensive data gathering and cleaning efforts to create a high-resolution database tracking price evolution across multiple products and supermarket chains, comprising over 700K data points.
- Performed extensive data wrangling and transformation in R to prepare the dataset for statistical modeling and analysis.
- Coded market structure indicators and coordination metrics, with reproducible code and methodology published via GitHub.
- Authored technical reports and research documentation for academic and applied audiences. Final document included in the Economic Department's research series.

Data Analyst, Ministry of Public Health – Montevideo, Uruguay

Nov 2018 to July 2019

- Supported regulatory officials overseeing healthcare providers and systems by delivering actionable data insights to improve access, quality, and accountability in healthcare services.
- Integrated and transformed over 200 datasets—including administrative records, survey responses, and qualitative inputs—using R, SQL, and Python to create robust performance metrics and access indicators.
- Developed interactive Business Intelligence (BI) dashboards in Tableau and QlikView to improve evaluation workflows, reduce manual processing, and improve reporting accuracy.
- Combined qualitative data from stakeholder workshops and interviews with quantitative analytics providing a comprehensive approach for the sector assessments and policy recommendations.
- Communicated actionable data insights to physicians, researchers, and government officials to support informed decision-making and policy development.

Senior Data Analyst, TCC – Montevideo, Uruguay

Mar 2017 to July 2019

- Led data science initiatives to optimize core business processes, reduce customer churn, and improve pricing strategies in competitive markets.
- Designed and implemented customer satisfaction surveys and retention analytics, to develop strategies that improved customer experience and loyalty.
- Deployed machine learning models on AWS and Azure to predict and reduce customer churn and identify potential default clients, resulting in a \$100K monthly reduction in operational costs and significant savings.
- Integrated structured and unstructured data from diverse sources—including Data Lake, Elasticsearch, Logs, MySQL, and MSSQL—to power comprehensive analytics and machine learning workflows.
- Led pricing strategy studies that informed pricing models and supported sustained growth in saturated markets
- Delivered periodic reports to senior management, including the CEO, investors, and regulators, to inform strategic decision-making.

External Consultant (Research Assistant), Inter-American Development Bank – Montevideo, Uruguay

Nov 2017 to Mar 2018, Nov 2016 to Mar 2017

- Contributed to the development of a cross-national dataset on government spending related to citizen security and justice systems across Latin America and the Caribbean.
- Conducted data validation and quality assessments across diverse public data sources, performing data cleaning and standardization to ensure consistency and comparability.
- Generated insights and visualizations that were featured in the IDB's flagship publication.

Analyst & Data Scientist, BCS- Consulting – Montevideo, Uruguay

Feb 2013 to July 2016

- Delivered tailored economic and market analytics for a range of clients, including banks, government agencies, and private firms, supporting strategic decision-making and operational efficiency.
- Developed and deployed predictive models for loan default risk, enabling banks to reduce lending risk while expanding access to housing finance for underserved customers.
- Integrated and cleaned over 1.5 million records from diverse sources to support a local government taxation initiative, resulting in increased revenue for green transportation incentive programs.
- Applied Erlang queueing models to optimize call center operations and reduce customer response times, improving operational performance and service quality.
- Worked independently as a lead analyst on multiple projects, managing data pipelines, model development, and stakeholder communications end to end.

- Designed and created data visualizations and interactive dashboards in Tableau to facilitate communication between clients and their stakeholders.

EDUCATION

- **Data Science Certificate** (2023)
DMAC, Iowa, United States
- **MS in Economics and International Finance** (2020)
Universidad de la República, Montevideo, Uruguay
- **Intensive English Program** (2020)
Iowa State University, Iowa, United States
- **Postgraduate Degree in Big Data Analysis** (2019)
Universidad ORT – Montevideo, Uruguay
- **BS in Economics** (2013)
Universidad de la República, Montevideo, Uruguay

SELECTED HONORS AND AWARDS¹

- **Love DATAWEEK2024**. First-place award for innovative visualizations. 10.25380/iastate.25429777
- **Provost's Innovation Award 2024** (Iowa State University) for advancements in statistical modelling.
- **Graduated with honors** and selected for the **President's List** (DMACC).
- **DMACC Foundation Scholarship** recipient.
- Inducted into **Phi Theta Kappa Honor Society**.
- Awarded **ORT University postgraduate scholarship**.
- **MS Dissertation distinction**. Dissertation included in the Economics Department's research series.

TEACHING EXPERIENCE

University of Montevideo, Montevideo, Uruguay

Master program in data analytics and data science

Data visualization and storytelling (2022- Present)

Introduction to Statistics (2023- Present)

Academic advisor for graduate dissertation Machine Learning projects (2023- Present)

BCS Academy, Montevideo, Uruguay

Professional training program

Introduction to Economics (2013-2016)

Introduction to Database and SQL (2013-2016)

MENTORING AND SERVICE EXPERIENCE

Women in IT (esp. Mujeres en IT): Mentor (2021-2022) and workshop instructor (2022) in a program aimed at empowering woman beginning their careers in Information Technology.

Include me (esp. Incluyeme): Mentor (2022-2023) in a program focused on improving employment readiness and career paths for marginalized communities.

¹ Distinctions awarded based on academic and professional performance.

Academic tutor volunteer.

ACSD tutoring program (2021): tutor for middle school level math.

Gurises Unidos (2016-2018): tutor and mentor for at-risk kids in Montevideo, Uruguay.

Un Techos para mi país (2006-2016): Tutor in the organization's educational programs implemented alongside their housing initiatives aimed at school-aged children in marginalized communities in Uruguay.